UAE AI in Education Policy Adviser

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Github link: https://github.com/BlackRock8888/ainquest-itu-policy-agent

This supplementary report presents AINQUEST's local policy-adviser prototype, a reproducible public demonstration and a separately licensed three-card UNESCO guidance dataset. The aim is to preserve evidence, scope and uncertainty in education-policy answers. Earlier research on 49 cards is reported separately because its source data and evaluation fixtures are absent from this delivery. The prototype is not a deployed legal or disciplinary service.

Supplement dated 12 November 2026. Code revision: 5773339f6d905d3c86d57f7238cf8eb47e4c420c. ITU acknowledged receipt of the earlier repository reference on 11 November; acceptance of the late entry is unconfirmed. Dataset paths and full SHA-256 hashes are recorded in the accompanying delivery manifest.

1. **Introduction**

   UAE students, teachers and administrators encounter AI guidance at national, emirate and institutional levels. The collected texts differ in scope, language and status; some available sources are announcements rather than full policy texts. A policy assistant must avoid mixing institutions and education levels, transferring foreign law into UAE cases, or treating detector scores as proof.

   Earlier research recorded 16 sources and 49 annotated cards spanning UAE education policy and global guidance. This supplement provides the public engine, the 16-source metadata index, one invented demonstration rule and three real UNESCO guidance excerpts. It does not recover the complete research corpus. English, Arabic and Russian answer frames preserve original-language evidence; native review remains outstanding. The agent makes no disciplinary decision.

2. **Description of the use case**

   The use case tests four risks: mixing institutions and education levels, confusing binding rules with guidance or foreign examples, answering from neighbouring documents when the requested source is missing, and treating detector signals as proof of misconduct. We have not evaluated all existing solutions and do not claim these risks occur in every product.

   The design separates scope extraction, BM25 retrieval and an evidence gate with four outcomes: answer, context_only, clarify and refuse. Source metadata records jurisdiction, level and binding status. This is an intended safeguard, not a guarantee: the separate-model test exposed inappropriate neighbouring-source answers. Detector and penalty questions are flagged for human review. Mandatory detector use and a prohibition on relying solely on detector output are compatible safeguards, not by themselves contradictory policies.

   The default kb/ directory contains one invented Demo University rule. kb_candidates/unesco/kb/ contains three English excerpts from Miao and Holmes,

Guidance for generative AI in education and research (UNESCO, 2023), printed pages 21 and 23. These are global, non-binding recommendations; attribution and CC BY-SA 3.0 IGO notices accompany them. The historical 49-card corpus remains separate. Its records reported 43 quotations matched against local snapshots and six without retained snapshots; those figures were not reproduced for this supplement.

3. **Use case requirements**
    - SRC: official sources and official institutional or vendor publications, with URL, publisher, fetch time and available hashes. Public access and short quotation length do not establish redistribution permission.
    - C / PP: source identifiers, jurisdiction, level, authority and version are recorded. Historic quotation and annotation checks require recovery and review. Enabled public logs preserve input queries; the earlier local privacy fix is not included. EN/RU/AR expansion has not received native-speaker review.
    - M: retrieval recall, decision accuracy and citation checks are measured separately. Optional local LLM rephrasing was not run. Its citation-ID check cannot establish that every generated claim is supported.
    - P: the intended policy is to answer only from applicable evidence, avoid cross-institution and cross-level transfer, and refuse unsupported questions. Baseline failures show that these controls are incomplete. Detector results are not used to make a disciplinary decision.
    - **D / SINK:** the answer frame is in the user's language and the quote stays in its original language. When --log is supplied, the CLI writes a decision record: run id, UTC time, model, corpus hash, retrieved, context and cited ids, gate reason, unknowns, terminal status, latency and cost.

4. **Mapping to ITU standards and domain policies**

    The node-to-dimension map derives from ITUAIReadiness commit 84010da9e9c443afcceeccb86d1700a662223395, with its MIT notice retained. This is our implementation mapping, not an ITU assessment. The final column describes historical research context; those policy texts are not all included in the delivered datasets.

    docs/standards-mapping.md now describes the published components and evidence limits. References to UAE AI Strategy and Charter cards are explicitly confined to the historical research corpus. Scope fields and multilingual output are implementation features, not readiness scores or proof of applicable law.

Node	Use case (our component)	Readiness 2.0 dims	Historical research context
SRC	Official UAE federal, emirate and institutional texts plus global examples, with provenance	1, 4, 13	UAE and Dubai AI ethics sources; distinct from the delivered UNESCO subset
C	Historical quotation collector: 43/49 snapshot matches reported, not reproduced here	1, 3, 13	–
PP	Scope, level and authority annotation; EN/RU/AR normalisation and glossary	1, 2, 3	MoE AI Literacy Framework 2026: data-protection duty for teachers

Node	Use case (our component)	Readiness 2.0 dims	Historical research context
M	BM25 retrieval plus evidence gate; optional local LLM (not run)	2, 3, 5, 9	UAE AI Charter: human oversight
P	Scope filter, authority metadata and human-review flags; baseline refusal failures remain	7, 8, 10	UoS Art. 10 (graduate scope; formal approval unverified), AUS, Turnitin: detector output not the sole basis; UAE and Dubai ethics: human override
D	Localised answer frame with original quotes and unknowns	5, 6, 12	UAE AI Ethics Guide: people informed of their interaction with AI
SINK	CLI/JSON for student, teacher or policy officer; optional decision log (--log)	5, 6, 12, 13	ADEK six safeguards: cognitive agency, human oversight
MLFO	Session and replay orchestrator for progressive events	5, 7, 8	–
Sandbox	Offline evaluation and replay harness. ITU sandbox not used.	10, 11	–

5. **Progressive scenarios and evidence**

- Step 1. An incomplete event arrives. The replay records a partial terminal status and produces no answer.
- Step 2. An English disclosure question at the invented Demo University is answered using the labelled synthetic rule.
- Step 3. A synthetic syllabus is stored as context. This demonstrates context handling, not interpretation of a real course policy.
- Step 4. Questions switch to Russian and Arabic while retaining institution scope and synthetic evidence. This is not a language-quality assessment.
- Step 5. An unsupported detector allegation arrives. With no applicable evidence in this dataset, the agent refuses.
- Step 6. The only rule is withdrawn. The next question is refused instead of reusing the withdrawn card. The six stages comprise eight event checks.

A separate UNESCO run loads the real guidance subset and asks one question in English, Russian and Arabic. All three outputs are context_only: original English quotations and source/page metadata appear as context examples, with no binding-policy citations. The guidance is not presented as UAE law. The historical UAEU research replay and its recorded 8/8 result are distinct and are not reproduced here.

6. Evaluation and limitations

Current public check	Result	Dataset	Meaning	Evidence
Existing smoke	2/2	1 invented rule	Selected mechanics	test_demo.py
Export regression	1/1	Synthetic source	Optional field handled	test_export.py
Assessment	6/6	6 authored cases	Expected outcomes	evals/
Progressive replay	8/8	8 authored events	State and withdrawal	evals/ scenarios/
UNESCO context	3/3	1 question in EN/RU/AR	context_only	check_unesco_ candidate.py

Public checks exercise selected mechanics after the authors inspected the implementation. They are not held-out tests, independent validation or an accuracy estimate. No external model, paid API, ITU sandbox or fine-tuning was used. The optional local LLM adapter remains untested.

Historical findings requiring a fresh audit

- Earlier research exposed answers drawn from neighbouring documents, unreliable aliases and institution/level inference, missing approval information and fallback after a policy version changed. Some weaknesses motivated targeted fixes, but the research fixtures and complete run files are absent here. A fresh assessment must retest these cases before real-policy reliability claims.

An export defect was reproduced in the public snapshot: a missing optional binding_evidence field caused KeyError. An explicit no-authority fallback and a regression check now handle that case without inventing authority. Exporting one synthetic file succeeds; ingestion by a deployed ITU pipeline has not been tested.

7. Policy and standards questions for validation

- Evidence access candidate. Earlier research found a reference to a classroom GenAI guide in the MoE AI Literacy Framework but did not retain a verified official full text of that guide. ADEK and SPEA entries were announcements. This is an evidence/coverage limitation, not proof that policies do not exist. Source-index identifiers: MOE_AI_LITERACY_FW_2026, ADEK_AI_LITERACY_2026 and SPEA_AI_ETHICS_FW_2026.
- Authority and version candidate. Evidence records should distinguish a draft, formal approval, effective date, exceptions and superseded clauses. The research note on the University of Sharjah graduate-policy document left approval unverified. We propose a testable record structure for this uncertainty, not a claim that no relevant standard exists. Source-index identifier: UOS_GRAD_AI_POLICY.
- Detector safeguards require complete scope. The earlier UAEU, University of Sharjah and Turnitin review described different safeguards. Requiring detector use and forbidding sole reliance on its result can coexist. We withdraw any inference of contradiction from those statements alone. Complete procedures, institutional scope and approval status require source review.

- Language and evidence transfer candidate. Quotations should retain page, institution, jurisdiction, version and binding status across language changes and handovers between agents. The glossary is not native-reviewed. We propose testing field loss with fixed scenarios; neither measured benefit nor a verified gap in an existing standard is claimed.
- Delegated action candidate. Our preparation process reached a late email without the report attachment. This motivates explicit records of who prepares, reviews and sends each version, the authorised action, deadline and receipt. It is an observed workflow failure and a proposed evaluation scenario, not evidence that a worldwide delegation standard is absent.
- Global reference boundary. The included UNESCO passages concern age-related safeguards and independent use of general-purpose GenAI. They retain their surrounding context as guidance. They establish neither a blanket school ban nor a UAE legal rule. Source: UNESCO 2023, printed pages 21 and 23, DOI 10.54675/EWZM9535.
- Publication boundary. The UNESCO-derived subset includes attribution, change notices and CC BY-SA 3.0 IGO terms. No full PDF, images or logo are included. Other research sources still require document-level reuse and substantive review. A metadata index of 16 sources is not evidence that all 16 have been ingested or approved.

Historical results from 9 November remain separate: development 20/20; previously exposed held-out 14/15; adversarial strict 3/20 and decision-only 9/20; safety regression 14/14; research replay 8/8; pytest 26 passed and 1 skipped. These were recorded in earlier local artifacts after targeted fixes. Their fixtures and outputs are not supplied or freshly reproduced here. Their denominators must not be combined with public checks; the 3/20 strict result remains a material limitation.

8. Next development milestones

Recover original research cards, fixtures, manifests and run outputs; assess rights and applicability source by source; review high-risk annotations with a domain specialist. Freeze a new test set before tuning. Measure scope selection, evidence support, justified refusal and language preservation separately, with source version and denominator.

Pilot one bounded education workflow with human review, exposing provenance, approval uncertainty and version changes. Seek ITU guidance on sharing validated standards questions with relevant Study Groups and developing a research contribution for Kaleidoscope. These are proposed next steps, not accepted activities.

References and reproduction: README, VALIDATION.md, docs/standards-mapping.md and kb/research-source-index.jsonl identify URLs, versions and commands. UNESCO: https://doi.org/10.54675/EWZM9535; licence: https://creativecommons.org/licenses/by-sa/3.0/igo/. The present work is not an official UNESCO publication and shall not be considered as such.